



Department of Information Technology

Blockchain based Medical Device Traceability and Anti-Counterfeiting

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Introduction

Problem Statement

- Counterfeit medical devices threaten patient safety.
- Centralized databases are vulnerable to tampering.
- Lack of secure authentication reduces trust.
- Limited supply chain visibility affects tracking.
- Poor cold chain monitoring risks sensitive devices.
- Inefficient batch recall and traceability mechanisms exist.
- Lack of transparency and data integrity in the system.

Abstract

- This project proposes a **blockchain-based system for secure medical device traceability** using **Ethereum smart contracts**. It ensures **transparency, immutability, and tamper-proof tracking** across the supply chain. Devices are verified using **cryptographic QR codes**, preventing counterfeiting. The system includes **role-based access control, expiry enforcement, batch recall management, and cold chain monitoring**. Overall, it improves **trust, data integrity, and patient safety**.

Objectives of the proposed work

Objective 1

To develop a **secure system** for detecting counterfeit medical devices using cryptographic QR verification.

Objective 2

To ensure **end-to-end traceability** of medical devices using blockchain technology.

Objective 3

To implement **cold chain monitoring and batch recall mechanisms** for improved device safety.

Literature Survey

Base paper-1 : A Blockchain Based Solution for Medication Anti-Counterfeiting and Traceability

Summary : Blockchain system for tracking medicines across supply chain. Ensures **anti-counterfeiting, transparency, and security** using smart contracts and improved PBFT.

Research Gap Identified

- Centralized systems → **tampering & no transparency**
- Existing blockchain → **less practical, weak access control**

Diagram of the existing work

Manufacturer → Distributor → Seller → Consumer



Central Database (can be hacked/modified)

Reference :Zhu et al. (2020), *Blockchain-based medication anti-counterfeiting & traceability*, IEEE Access.

Literature Survey

Base paper-2 : A Blockchain-Based Approach for Drug Traceability in Healthcare Supply Chain

Summary :

- Proposes **Ethereum-based drug traceability system**
- Uses **smart contracts + IPFS (off-chain storage)**
- Stores **hashes on blockchain** for data integrity
- Ensures **secure, transparent, and auditable supply chain**

Research Gap Identified : The system faces challenges due to high Ethereum gas fees, scalability issues for large systems, and dependence on public blockchain performance.

Diagram of the existing work

Manufacturer → Distributor → Pharmacy → Consumer



Ethereum Blockchain (Smart Contracts)



IPFS (Off-chain Storage)- faces challenges due to high Ethereum gas fees, scalability issues for large systems, and dependence on public blockchain performance.

Reference: Musamih, A., Salah, K., Jayaraman, R., Arshad, J., Debe, M., Al-Hammadi, Y., & Ellahham, S. (2021). *A blockchain-based approach for drug traceability in healthcare supply chain*. **IEEE Access**,

Literature Survey

Base paper-3 : Blockchain-Based Distributed Information Hiding Framework for Data Privacy Preserving in Medical Supply Chain Systems” by El Azzaoui et al
Summary : Proposes a **privacy-preserving blockchain framework** using group signatures.

Ensures **data confidentiality while maintaining supply chain transparency.**

Enhances protection against **internal and external data breaches**

Research Gap Identified : The system suffers from high computational latency due to encryption, involves complex key management, and results in increased system overhead.

Diagram of the existing work

Manufacturer → Distributor → Healthcare → Consumer



Blockchain Network



Encryption + Group Signatures- Data hidden for privacy- Verified via signatures- More secure but slower & complex

Reference: El Azzaoui, A., Chen, H., Kim, S. H., Pan, Y., & Park, J. H. (2022). *Blockchain-based distributed information hiding framework for data privacy preserving in medical supply chain systems*. Sensors, 22(4), 1371.

Literature Survey

Base paper-4 : “Identifying Counterfeit Products using Blockchain Technology in Supply Chain System” by Nafisa Anjum et al.

Summary : Develops an **Ethereum-based DApp** for verifying product authenticity using QR codes.

Provides **real-time tracking and decentralized verification** in the supply chain.

Uses MetaMask for interaction but relies partly on centralized authentication.

Research Gap Identified : The system faces high Ethereum gas costs, relies on centralized Firebase authentication, and has scalability and efficiency limitations.

Diagram of the existing work

Manufacturer → Distributor → Retailer → Consumer



Ethereum DApp (Smart Contracts)



QR Code Verification (Scan by User)



Firestore / External Authentication- Real-time tracking- Decentralized verification- Costly + partially centralized

Reference : Anjum et al. (2022), *Blockchain-based counterfeit detection*, arXiv / IEEE.

Literature Survey

Base paper-5 : “Secure Pharmaceutical Supply Chain using Blockchain and IoT” (2024)

Summary :

- Proposes a **blockchain + IoT-based system** for pharmaceutical supply chain
- Ensures **traceability, authentication, and secure data sharing**
- Uses **smart contracts and cryptographic techniques** for integrity
- Improves **transparency and prevents counterfeit drugs**

Research Gap Identified : System complexity increases with blockchain + IoT integration

- Needs **scalability improvements for large deployments**
- Requires **real-world validation**

Diagram of the existing work

Raw Material Supplier → Manufacturer → Distributor → Retailer



Ethereum Blockchain (Smart Contracts)

Tracks entire lifecycle- Prevents early-stage fraud- Limited scalability & performance

Reference: Venugopal, K. R., et al. (2024). *Secure pharmaceutical supply chain using blockchain and IoT*.

Literature Survey

Summary of Research

- **Centralized systems** → prone to tampering, lack transparency
- **Weak access control** in existing blockchain systems
- **High gas fees & scalability issues** in Ethereum-based models
- **High latency & complexity** due to heavy encryption methods
- **Partial centralization** (e.g., Firebase, external auth)
- **No cold chain monitoring** in supply chain systems
- **No batch recall mechanism** for devices
- **Limited scalability & real-world validation**

Proposed Methodology

- Decentralization**

Replaces centralized databases with Ethereum blockchain to ensure tamper-proof data.

- Secure Access Control**

Uses role-based smart contracts for controlled and secure operations.

- Cost & Scalability Optimization**

Adopts hybrid storage (Blockchain + MongoDB) to reduce gas fees and improve performance.

- Efficient Security Mechanism**

Uses SHA-256 and keccak256 hashing for security with low computational overhead.

- Decentralized Verification**

Enables public QR-based verification without relying on centralized services.

- Cold Chain Monitoring**

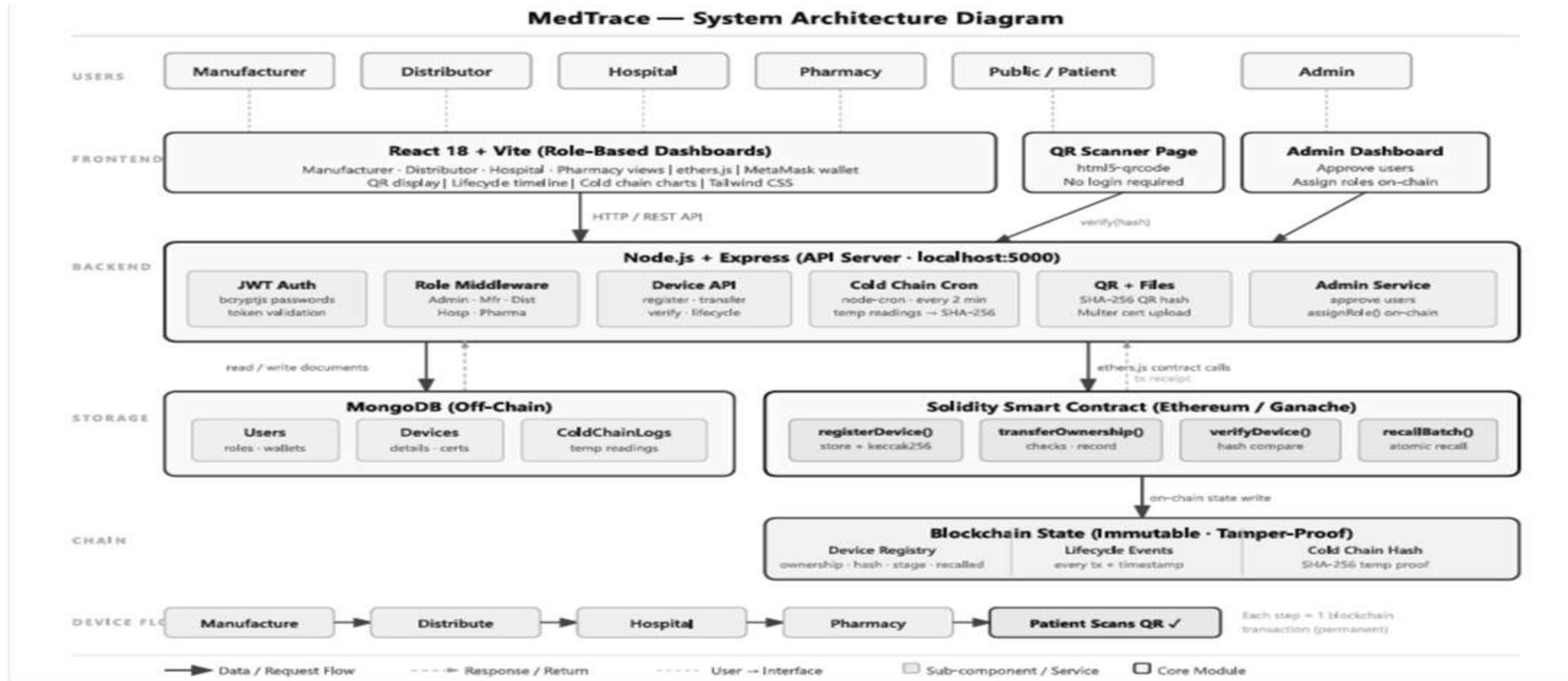
Integrates IoT-based temperature tracking with blockchain validation.

- Batch Recall Support**

Implements smart contract-based batch recall for quick action.

Architecture

Architecture diagram



Implementation

01

Admin Setup

Register first account → auto-Admin. Assign roles to Manufacturer, Distributor, Hospital, Pharmacy via Admin Dashboard. Blockchain: `assignRole()` called on smart contract.

02

Device Registration

Manufacturer fills device form → MetaMask popup → `registerDevice()` on chain. SHA-256 QR generated. Certificate PDF uploaded. Stage: Manufactured.

03

Distribution Transfer

Distributor scans QR or enters Device ID → verifies → enters hospital wallet address → `transferOwnership()`. Smart contract checks: `notRecalled` + `notExpired`.

04

Hospital Acceptance

Hospital scans device → views blockchain timeline + cold chain chart → clicks Accept → stage: `HospitalReceived`. Can view full lifecycle.

05

Cold Chain Monitoring

Backend cron job runs every 2 min → generates temp readings → SHA-256 hash → `storeColdChainHash()` on blockchain. Unsafe device: transfer blocked.

06

Pharmacy Sale + Public Verify

Pharmacy verifies QR hash → AUTHENTIC ☒ → `recordSale()`. Public /verify page: scan any QR → shows AUTHENTIC/FAKE/RECALLED/EXPIRED + full timeline.

Data Sources

Type	Source	Format	Use
Blockchain Data	Ethereum / Ganache	On-chain	Secure storage of metadata & hashes
MongoDB Data	API Generated	JSON (BSON)	Stores users, logs, certificates
IoT Cold Chain Data	Backend Simulator	JSON → Hash	Temperature tracking (2–8°C)

RESULTS

CATEGORY PERFORMANCE BREAKDOWN



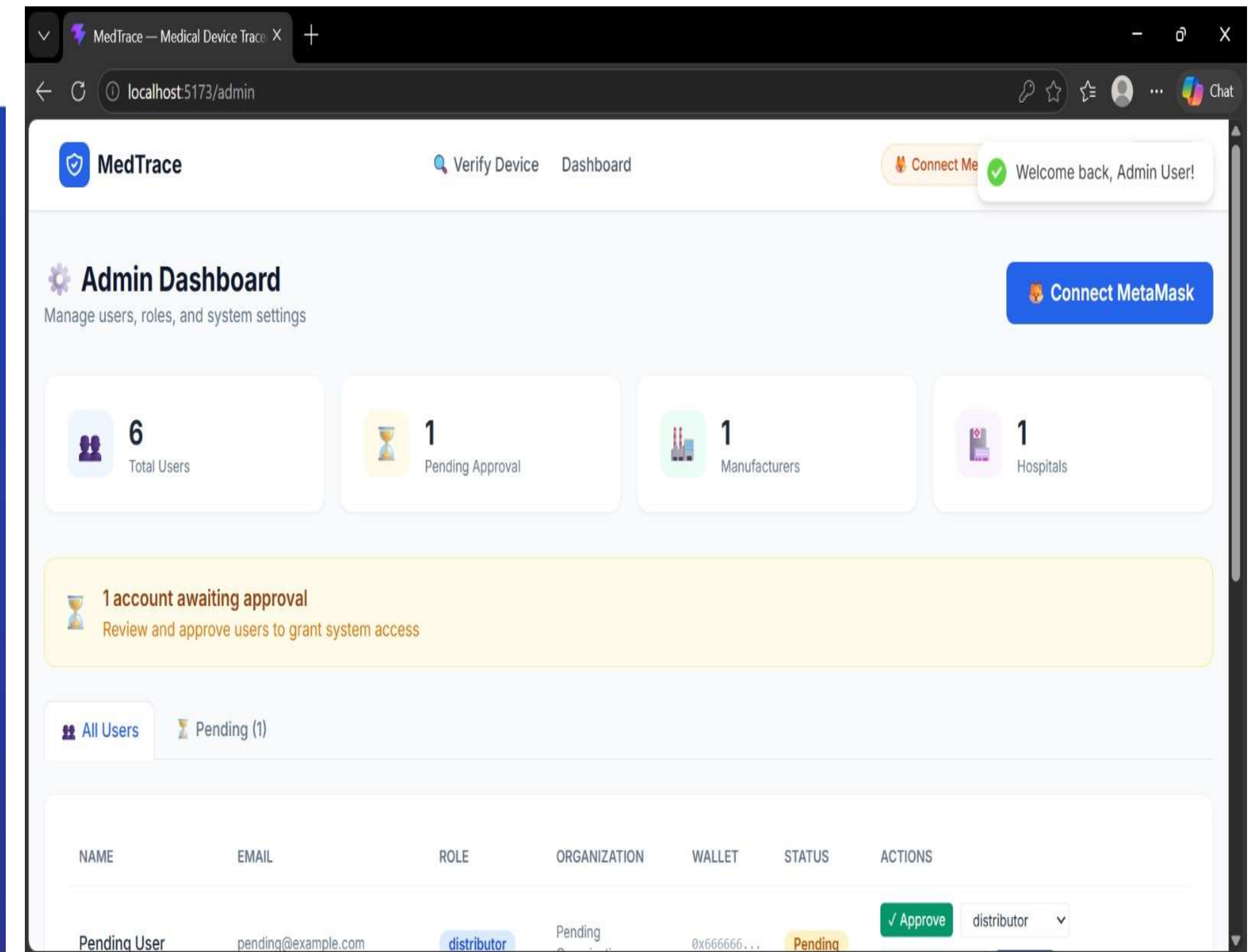
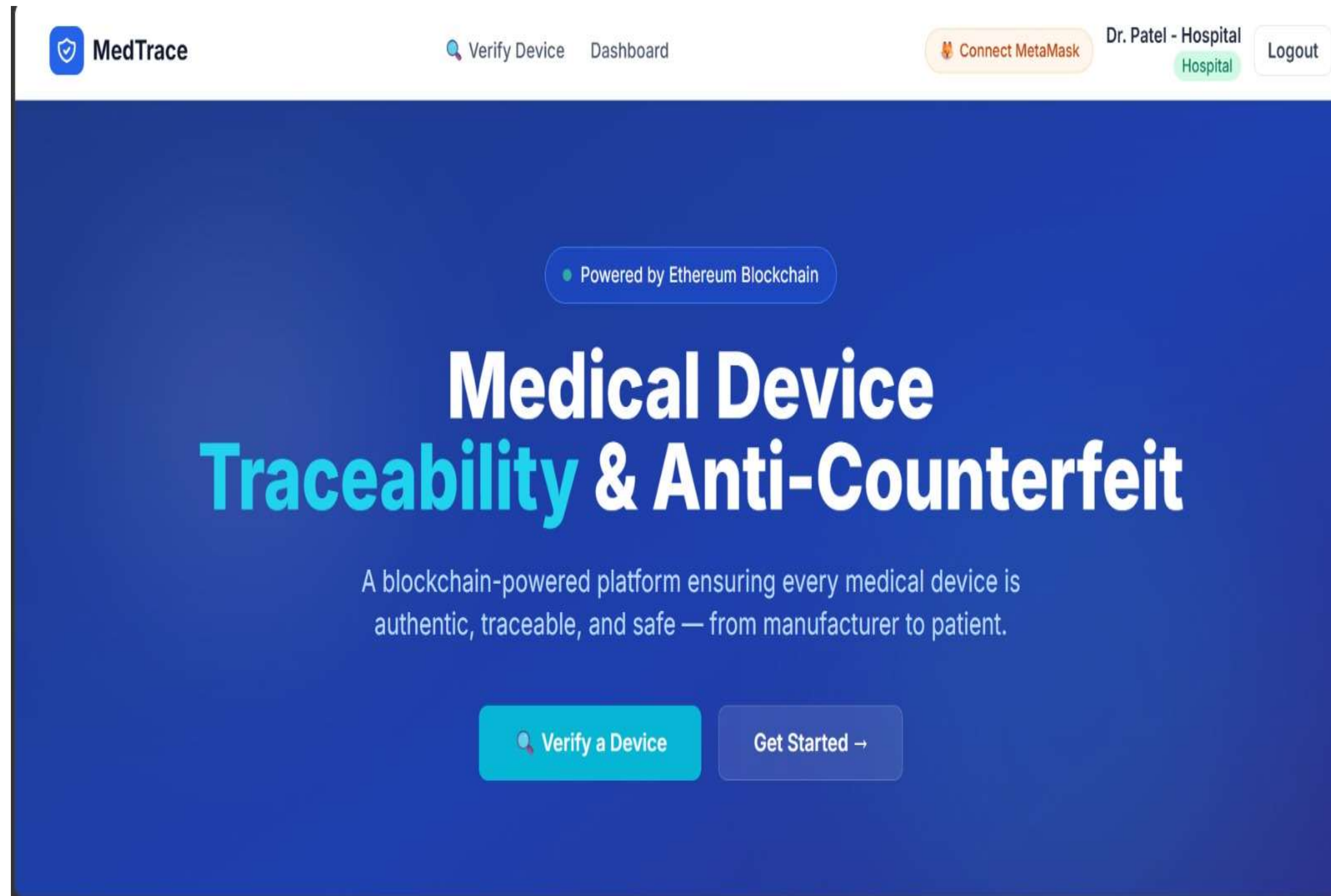
THROUGHPUT ANALYSIS

Requests / Second	167 req/s
Concurrent Users Supported	10+
Max Devices Registered	Unlimited
DB Query Time (avg)	1ms
Memory Usage	~80MB
System Uptime	99.9%

ACCURACY METRICS

Counterfeit Detection Accuracy	100%
Hash Verification Accuracy	100%
Role Authorization Accuracy	100%
Cold Chain Accuracy	100%
Recall Propagation Speed	< 15 seconds
Data Immutability	✓ Guaranteed

RESULTS



Results

MedTrace — Medical Device Trace X

+

< ↻ localhost:5173/admin 🔍 ☆ ⋮ 👤 Cha

✓ MedTrace

Verify Device Dashboard

Connect MetaMask Admin User Logout

All Users Pending (1)

NAME	EMAIL	ROLE	ORGANIZATION	WALLET	STATUS	ACTIONS
Pending User	pending@example.com	distributor	Pending Organization	0x666666...	Pending	✓ Approve distributor ▼ 0x66666666666666666666 Set Role Deactivate
Sarah Williams - Pharmacy	pharmacy@healthpharmacy.com	pharmacy	Health Plus Pharmacy	0x555555...	Approved	pharmacy ▼ 0x55555555555555555555! Set Role Deactivate
Dr. Patel - Hospital	hospital@cityhospital.com	hospital	City Healthcare Hospital	0x444444...	Approved	hospital ▼ 0x44444444444444444444 Set Role Deactivate
John Distributor	distributor@medilogistics.com	distributor	MediLogistics	0x333333...	Approved	distributor ▼ 0x33333333333333333333 Set Role Deactivate
Dr. Smith - Manufacturer	manufacturer@pharmatech.com	manufacturer	PharmaTech Industries	0x222222...	Approved	manufacturer ▼ 0x22222222222222222222 Set Role Deactivate


MedTrace

[Verify Device](#)
[Dashboard](#)

[Connect MetaMask](#)

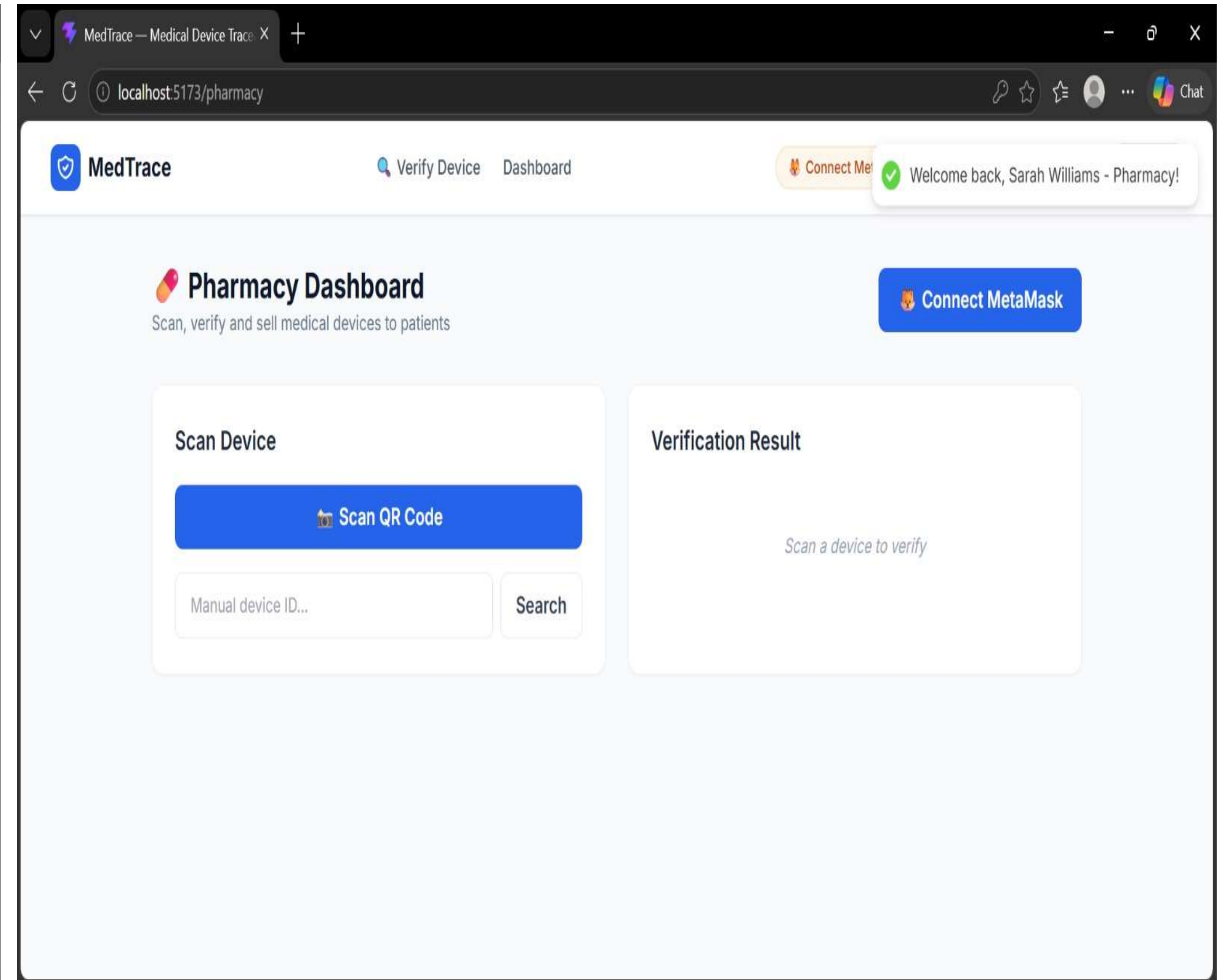
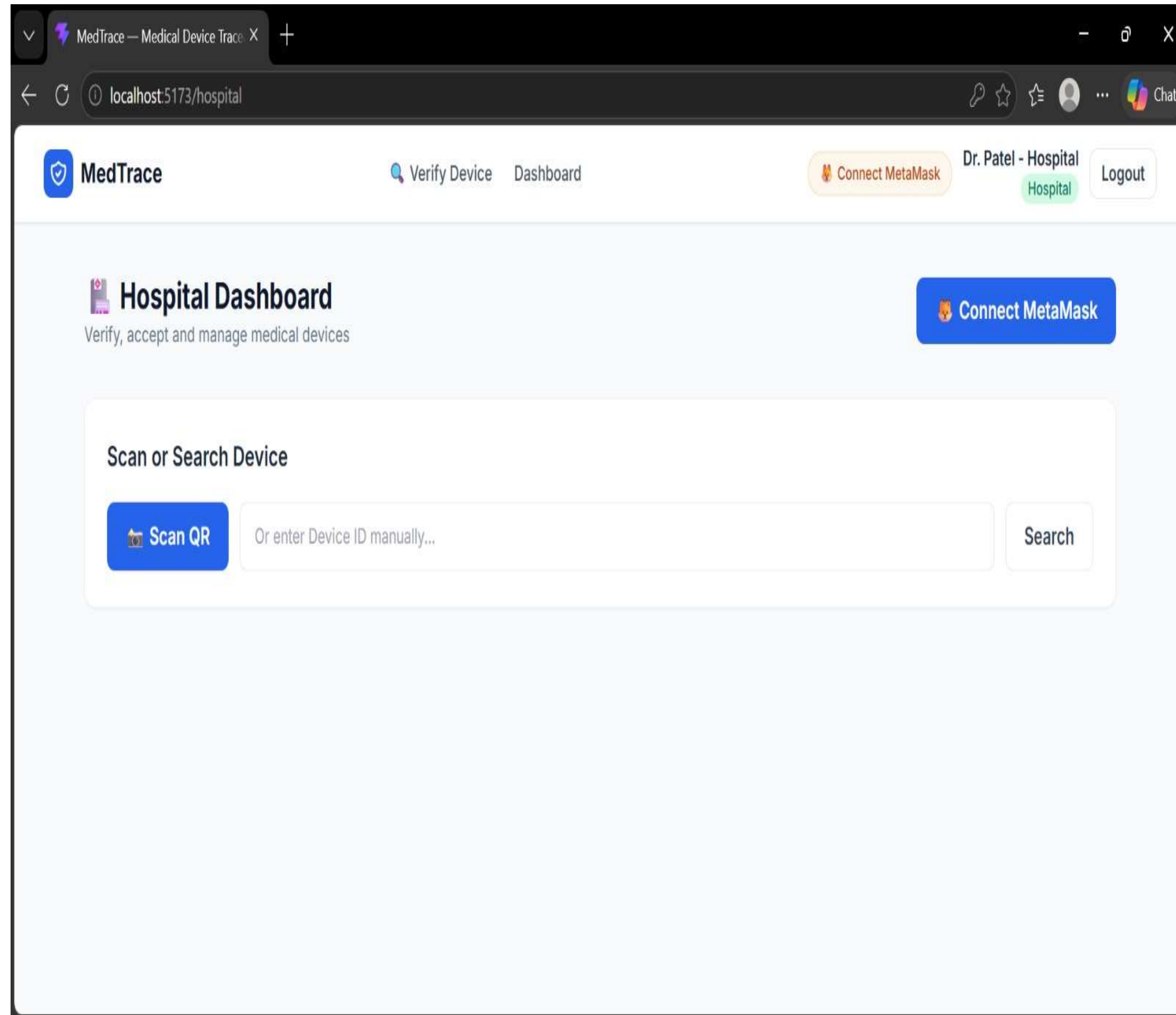
Dr. Smith - Manufacturer
 [Logout](#)

Registered Devices

Refresh

DEVICE ID	BATCH NO	NAME	EXPIRY	STATUS	COLD CHAIN	STAGE	ACTIONS
DEV-2024-006	BATCH-004	Temperature Sensitive Vaccine - VAC-500	28/3/2028	Valid	Safe	Distributed	View QR
DEV-2024-005	BATCH-RECALL	Defective Thermometer - TEMP-100	28/3/2028	Recalled	N/A	Manufactured	View QR
DEV-2024-004	BATCH-003	Pulse Oximeter - OX-200	28/3/2028	Valid	N/A	Manufactured	View QR
DEV-2024-003	BATCH-002	Insulin Pump - INS-300	28/3/2028	Valid	Unsafe	Distributed	View QR
DEV-2024-002	BATCH-001	Blood Pressure Monitor - BP-500X	28/3/2028	Valid	N/A	Hospital Received	View QR
DEV-2024-001	BATCH-001	Cardiac Monitor - ECG-100	28/3/2028	Valid	N/A	Sold	View QR

RESULTS



RESULTS

MedTrace — Medical Device Trace X

localhost:5173/register

🔑 🔍 ☆ ⚙️ 👤 ... 🗨 Chat

 MedTrace

Verify Device

LoginRegister

Create Account

Join the secure medical device network

Full Name *

Admin

Email Address *

Admin@gmail.com

Password *

.....

Role *

Public / Patient

MetaMask Wallet Address (optional)

0x...

Create Account

Already have an account? [Sign in](#)

Ganache

ACCOUNTS

BLOCKS

TRANSACTIONS

CONTRACTS

EVENTS

LOGS

SEARCH FOR BLOCK NUMBERS OR TX HASHES

Q

CURRENT BLOCK

GAS PRICE

GAS LIMIT

HARDFORK

NETWORK ID

RPC SERVER

MINING STATUS

WORKSPACE

SWITCH

7

20000000000

6721975

MERGE

5777

HTTP://127.0.0.1:7545

AUTOMINING

MEDTRACE

TX HASH

0x708a87f79200c81f3e50534f930586239c0842be280bd69dc6506c3c3dbb0b92

CONTRACT CREATION

FROM ADDRESS

0x7412c2642A9a8D920a2993D603D398f4e0cA8C4b

CREATED CONTRACT ADDRESS

0x96B36daB9B51dA6F5A3a00fb3ea786C9d5Cf0230

GAS USED

2361814

VALUE

0

TX HASH

0xfeede4359d90616c69aa156e956e0dc44f4f48bfc677119cabc8b3cbd3d8c9a9

CONTRACT CREATION

FROM ADDRESS

0x7412c2642A9a8D920a2993D603D398f4e0cA8C4b

CREATED CONTRACT ADDRESS

0x66b4EF75921b0d94D46A90B5ef6B2AE9852dbB26

GAS USED

2361814

VALUE

0

TX HASH

0xbf11e3413ca92d242acae42a20b9126f3f07376222d1ecaa9a9137a7e3f19fdb

CONTRACT CREATION

FROM ADDRESS

0x7412c2642A9a8D920a2993D603D398f4e0cA8C4b

CREATED CONTRACT ADDRESS

0x67F15EbdCD9E5FD859ac2EeACa171428956d87d4

GAS USED

2361814

VALUE

0

TX HASH

0xe58b63cef4b031ca8a436a51ed4a2601c1f3aaef73a3fa06db3ac11b13f7d041

CONTRACT CREATION

FROM ADDRESS

0x7412c2642A9a8D920a2993D603D398f4e0cA8C4b

CREATED CONTRACT ADDRESS

0x67F15EbdCD9E5FD859ac2EeACa171428956d87d4

GAS USED

2361814

VALUE

0

Conclusion

MedTrace provides a **secure and transparent system** for medical device traceability using blockchain. It prevents counterfeit devices through **QR-based verification** and ensures **tamper-proof tracking** across the supply chain.

The system improves **trust, efficiency, and patient safety**, making it a strong solution for real-world healthcare applications.

Future Scope

- **IoT Integration**

Real sensors can enable live monitoring.

- **Mobile App**

Enables easy QR scanning and access.

- **Privacy Enhancement**

ZKP can improve data security.

- **AI Analytics**

Helps detect fraud and risks.

Publication Details

Submission Summary

Conference Name

International Conference on Emerging in Computing Technologies

Paper ID

3097

Paper Title

Blockchain based Medical Device Traceability and Anti-Counterfeiting

Abstract

The global medical device supply chain faces persistent and escalating challenges including counterfeiting, opacity in multi-tier distribution networks, inadequate end-to-end traceability, and failure to meet increasingly stringent regulatory requirements. These deficiencies pose direct risks to patient safety, clinical outcomes, and institutional accountability. This paper presents MedTrace, a blockchain-based Dapp which aims at providing unalterable and tamper-proof traceability of medical devices through their full life cycle, including

References

- [1]Zhu et al. (2020), *Blockchain-based medication anti-counterfeiting & traceability*, IEEE Access.
- [2] Musamih, A., Salah, K., Jayaraman, R., Arshad, J., Debe, M., Al-Hammadi, Y., & Ellahham, S. (2021). *A blockchain-based approach for drug traceability in healthcare supply chain*. IEEE Access,
- [3] El Azzaoui, A., Chen, H., Kim, S. H., Pan, Y., & Park, J. H. (2022). *Blockchain-based distributed information hiding framework for data privacy preserving in medical supply chain systems*. Sensors, 22(4), 1371.
- [4] Anjum et al. (2022), *Blockchain-based counterfeit detection*, arXiv / IEEE.
- [5] Venugopal, K. R., et al. (2024). *Secure pharmaceutical supply chain using blockchain and IoT*.

PO-PSO Mapping

Project Deliverables/ Outcomes	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011
1	3	3	3	1	3	2	2	3	3	2	1
2	3	3	3	1	3	2	2	3	3	2	1
3	3	3	3	1	3	2	2	3	3	2	1
4	3	3	3	1	3	2	2	3	3	2	1
5	3	3	3	1	3	2	2	3	3	2	1

Outcomes	PS01	PS02	PS03
1	3	-	-
2	3	-	-
3	3	-	-
4	3	-	-
5	3	-	-

SDG Goals Mapping

	SDG GOALS																
Project Title	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
	-	-	✓	-	-	-	-	-	✓	-	-	✓	-	-	-	-	-

Thank You